Hostel Management System

Code Coverage Report

Statement coverage achieved by the automated test suite

Project	Hostel Management System (HMS)
Team	Team 3
Authors	Meghana P (PES2UG24CS642) B S Sharath Chandra Reddy (PES2UG24CS630) Adarsha V (PES2UG24CS621) Mohammadnoor Sirasgi (PES2UG25CS812)
Repository	github.com/adarshv15/hostel-management-system-team3 (commit 857e9c1)
Baseline document	Software Requirements Specification v1.0 (TEAM_3_HMS_SRS)
Report version / date	1.0 / 09-11-2026
Status	Draft for team review
Test period	Testing phase (plan: 2 Nov to 6 Nov); this run executed 09-11-2026

1. Introduction

This report gives the statement (line) coverage achieved by the automated test suite on the Hostel Management System at commit `857e9c1`. Statement coverage answers one question: which executable lines of the application ran at least once while the 32 unit tests ran. It does not show whether the tests check the right results; that is covered by the Unit Test Report.

Branch coverage, which measures whether both outcomes of each decision were taken, is reported separately in the Branch Coverage Report.

2. Measurement Method

Item	Detail
Operating system	Linux 6.18 (cloud sandbox)
Python	3.13.18 (the project CI targets Python 3.12)
Django	5.2.19 (requirements.txt allows >=5.1,<6.0)
Coverage tool	coverage.py 7.17.2 with `--branch`
Database	SQLite, Django-managed in-memory test database
Command	`coverage run --branch --source=hostel,hms_project manage.py test -v 2`
Reports	`coverage report -m`, `coverage json`, `coverage html`
Scope	All Python files in `hostel/` and `hms_project/`. The test module `hostel/tests.py` is excluded from the figures below, because tests that run themselves would inflate the result. Empty `__init__.py` files are skipped.
Execution date	09-11-2026

3. Summary of Results

Measure	Result
Executable statements	321
Statements executed	278
Statements missed	43
Statement coverage	86.6% (278 of 321)
Test result	32 of 32 tests passed

Result: statement coverage is **86.6%**. The two core application files, `models.py` (93.5%) and `admin.py` (91.2%), are above 90%. Three files account for 25 of the 43 missed statements: `signals.py` (17, a module that is never loaded) and `asgi.py` and `wsgi.py` (4 each, deployment entry points that a test client does not run).

The Test Plan's coverage target was not available in the repository. If it sets a threshold, compare it with the figure above.

3.1 Coverage by file

File	Statements	Executed	Missed	Coverage
`hostel/models.py`	155	145	10	93.5%
`hostel/admin.py`	91	83	8	91.2%
`hostel/signals.py`	17	0	17	0.0%
`hostel/views.py`	8	8	0	100.0%
`hostel/apps.py`	4	4	0	100.0%
`hostel/migrations/0001_initial.py`	9	9	0	100.0%
`hms_project/settings.py`	25	25	0	100.0%
`hms_project/urls.py`	4	4	0	100.0%
`hms_project/asgi.py`	4	0	4	0.0%
`hms_project/wsgi.py`	4	0	4	0.0%
Total	**321**	**278**	**43**	**86.6%**

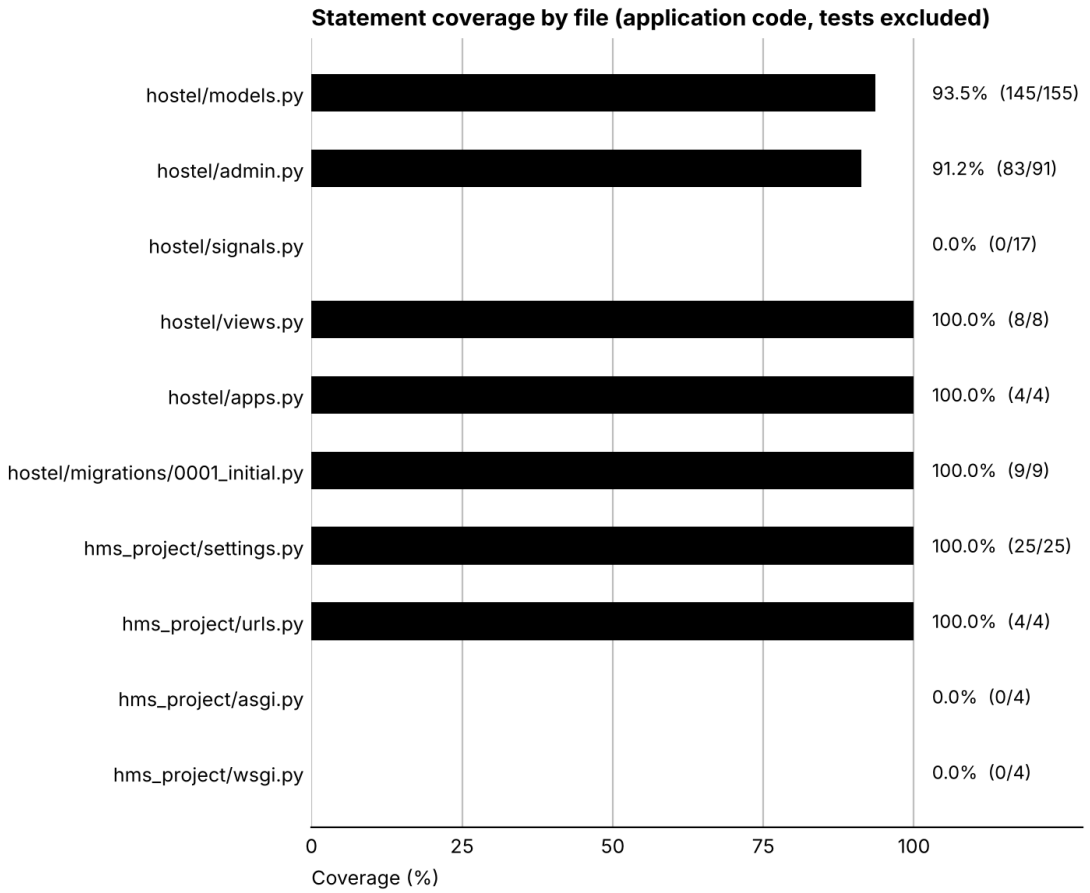


Figure 1. Statement coverage for each application file.

3.2 Sensitivity to scope

Scope	Statements	Missed	Coverage
Application code, tests excluded (headline)	321	43	86.6%
Same, also excluding `asgi.py` and `wsgi.py` (entry points)	313	35	88.8%
Including `hostel/tests.py` (raw coverage.py output)	526	43	91.8%

The headline figure is the first row. The other rows are shown so the effect of scope choices is visible.

4. Uncovered Code

All 43 missed statements, grouped by cause and ranked by what they put at risk.

Location	Lines	Code	Risk	Cause
`hostel/models.py`	62	`Allocation.clean`: raise "Only active students can be allocated a bed"	Medium	Business rule with no test
`hostel/admin.py`	152, 154, 159, 169, 173, 174	`FeeInvoiceAdmin`: annotated queryset, `total_paid`, `balance_due` columns	Medium	Admin list never rendered in tests
`hostel/admin.py`	45, 58	`delete_model`: audit event and delete for a single record	Low	Single delete untested (bulk is tested)
`hostel/signals.py`	2 to 80 (17 statements)	Audit signal receivers and registration	Medium	Module never imported; no `ready()` in `apps.py`

Location	Lines	Code	Risk	Cause
`hostel/models.py`	71, 84, 115, 130, 149, 166, 180, 191, 206	`__str__` of Allocation, FeeInvoice, PaymentRecord, Complaint, LeaveRequest, VisitorLog, MealAttendance, Notice, AuditEvent	Low	Display strings not asserted
`hms_project/asgi.py`	1 to 4	ASGI application entry point	Low	Not run by the test client
`hms_project/wsgi.py`	1 to 4	WSGI application entry point	Low	Not run by the test client

Missed statements by file: `models.py` 10, `admin.py` 8, `signals.py` 17, `asgi.py` 4, `wsgi.py` 4. Total 43.

5. Recommendations

Five small tests would remove every missed statement outside `signals.py` and the entry points:

1. Validate an allocation for a Pending, Rejected and Inactive student and assert it is rejected (covers `models.py` line 62).
2. Open the invoice admin changelist after recording a part payment and assert the paid and balance values (covers `admin.py` 152 to 174).
3. Delete one record through the admin and assert one DELETE audit event (covers `admin.py` 45 and 58).
4. Assert the string form of each model (covers the nine `__str__` lines).
5. Decide on `signals.py`: register it in `HostelConfig.ready()` and test it (taking care not to log twice alongside `AuditLoggingAdmin`), or delete it.

Covering the 18 missed statements in `models.py` and `admin.py` would raise statement coverage from 86.6% to **92.2%** (296 of 321). Resolving `signals.py` as well, by testing it, would reach 97.5% (313 of 321), leaving only the two entry-point files.

6. Limitations

- Statement coverage shows that a line ran, not that its result was checked. Several covered lines are exercised by tests that only create records.
- It says nothing about code that does not exist. Requirements that are not implemented (see the System Validation Report) are not visible here.
- Figures come from one run on Python 3.13.18. Results on the project's CI (Python 3.12) are expected to match but were not measured.
- Template rendering and Django internals are not measured.

7. How to Reproduce

```
python -m pip install -r requirements.txt
coverage erase
coverage run --branch --source=hostel,hms_project manage.py test
coverage report -m --omit="*/tests.py"
coverage html --omit="*/tests.py"   # open htmlcov/index.html
```

8. Conclusion

The test suite executes 86.6% of the application's statements. The business logic in `models.py` and `admin.py` is well covered; the gaps are one untested business rule, one untested admin display path and an unused signals module, all of which are quick to close.

Sign-off

Role	Name	Signature	Date
Prepared by			
Reviewed by			
Approved by (Project Guide)			

Appendix A. Raw coverage.py Output

Output of `coverage report -m --skip-empty` including `hostel/tests.py` (the `Cover` column here combines statements and branches, as coverage.py does when `--branch` is on).

```
Name                                    Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------
hms_project/asgi.py                         4      4      0      0     0%   1-4
hms_project/settings.py                    25      0      0      0   100%
hms_project/urls.py                         4      0      0      0   100%
hms_project/wsgi.py                         4      4      0      0     0%   1-4
hostel/admin.py                            91      8      2      1    90%   45-58, 81->84, 152-159,
169, 173-174
hostel/apps.py                              4      0      0      0   100%
hostel/migrations/0001_initial.py           9      0      0      0   100%
hostel/models.py                          155     10     20      4    92%   62, 63->exit, 71, 84,
100->exit, 115, 130, 145->exit, 149, 166, 180, 191, 206
hostel/signals.py                          17     17      4      0     0%   2-80
hostel/tests.py                           205      0      2      0   100%
hostel/views.py                             8      0      0      0   100%
-----------------------------------------------------------------------------------
TOTAL                                     526     43     28      5    91%

3 empty files skipped.
```